

**SOFTWARE TESTING**  
**MODULE END ASSESSMENT 3**  
**JAVA**

### Problem 1

Write a Java program to calculate the factorial of a given positive integer.

#### Program:

```
package JavaXam;

import java.util.Scanner;

public class FactorialPrgm {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        for(int j = 1; j<=3; j++) {

            System.out.print("\nEnter a positive integer: ");

            int N = sc.nextInt();

            long factorial = 1;

            if(N == 0) {

                factorial = 1;

            } else {

                for(int i = 1; i<=N; i++) {

                    factorial = factorial*i;

                }

            }

            System.out.println("Factorial of " + N + " is " + factorial);

        }

        sc.close();

    }

}
```

**Program Output:**

Enter a positive integer: 5

Factorial of 5 is 120

Enter a positive integer: 0

Factorial of 0 is 1

Enter a positive integer: 7

Factorial of 7 is 5040

**Explanation:**

- The program calculates the factorial of a positive integer entered by the user.
- 'N' variable is used to store the input number.
- Used 'factorial' variable to store the factorial result.
- Used for loop to multiply all numbers from 1 to N.
- If the input is 0, the program returns 1 because  $0! = 1$ .
- The long data type is used for the factorial variable to handle larger values.

## Problem 2

Write a Java program to generate the Fibonacci series until the sequence reaches or exceeds the limit of 50.

### Program:

```
package JavaXam;

public class FibonacciPrgm {

    public static void main(String[] args) {

        int first = 0;

        int second = 1;

        System.out.print("Output: " + first + ", " + second);

        while(true) {

            int next = first + second;

            if(next>50) {

                break;

            }

            System.out.print(", " + next);

            first = second;

            second = next;

        }

    }

}
```

### Program Output:

Output: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34

**Explanation:**

- The program generates the fibonacci series starting from 0 and 1.
- It uses a while loop to continue generating numbers until the next value exceeds 50.
- Stored the 0 and 1 in 'first' and 'second' variables.
- Stored the next number in the sequence in 'next' variable.
- Each new number is calculated as the sum of the previous two numbers.
- The program stops when the next fibonacci number becomes greater than 50, ensuring the sequence does not exceed the specified limit.